WEEK 2 · 16–19 NOVEMBER 2026

Material discrimination and target detection

What is this pixel made of — and what is hiding inside it?

Week 2 · Session 3 · practical: 14_material_discrimination_target_detection.ipynb

LEARNING OBJECTIVES

What you will be able to do

- Compare spectra with **SAM** and **SID**, and explain why both ignore brightness
- Build rule images, choose thresholds and produce a SAM classification map
- Use **continuum removal** to match diagnostic absorption features rather than whole spectra
- Derive the **matched filter**, **ACE** and **CEM** from background statistics
- Benchmark detectors honestly with a target-implantation experiment and ROC curves
- Unmix a pixel with **N-FINDR** endmembers and **FCLS** abundances, and read the residual map
- Explain why methane retrieval uses a matched filter on radiance, not a band ratio

1. Three questions, three families

Choosing the right tool before the right code

WHY THIS SESSION EXISTS

A multispectral sensor tells you a pixel is bright or green.
A hyperspectral sensor can tell you what it is made of.

And often whether a material you are looking for is hiding inside a 30 m pixel that is mostly something else.

Match the method to the question

Family	Question it answers	Methods
Spectral similarity	Which reference does this whole pixel look like?	SAM, SID, feature matching
Target detection	Is a small amount of material X present?	Matched filter, ACE, CEM
Spectral unmixing	What fraction of the pixel is each material?	N-FINDR / VCA + NNLS / FCLS

2. Spectral similarity

Comparing shapes, not brightness

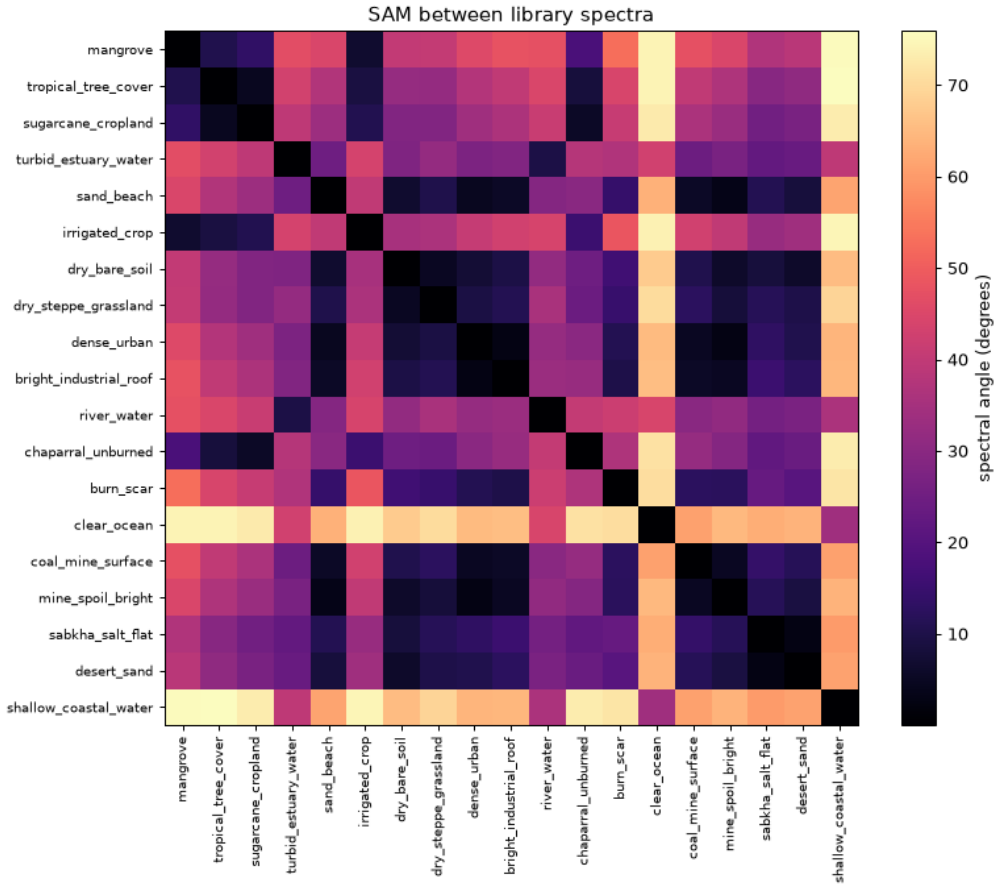
Three ways to compare two spectra

- **Euclidean distance** $\|x - r\|$ — sensitive to brightness *and* shape
- **SAM** (Kruse et al., 1993): the angle between the two vectors, $\alpha = \arccos(x \cdot r / (\|x\|\|r\|))$
  - Multiplying a spectrum by a constant does not change the angle — shape only
- **SID** (Chang, 2000): treat each spectrum as a probability distribution over bands
  - Symmetric Kullback–Leibler divergence — weighs *where* the energy sits
- Both SAM and SID are **brightness-invariant**; Euclidean distance is not

Brightness invariance, measured

Comparison	Euclidean	SAM (rad)	SID
Dry bare soil vs the same soil at 50 % illumination	3.837	0.000	0.000
Dry bare soil vs dry steppe grassland	0.948	0.084	0.010

Which materials can be told apart at all?



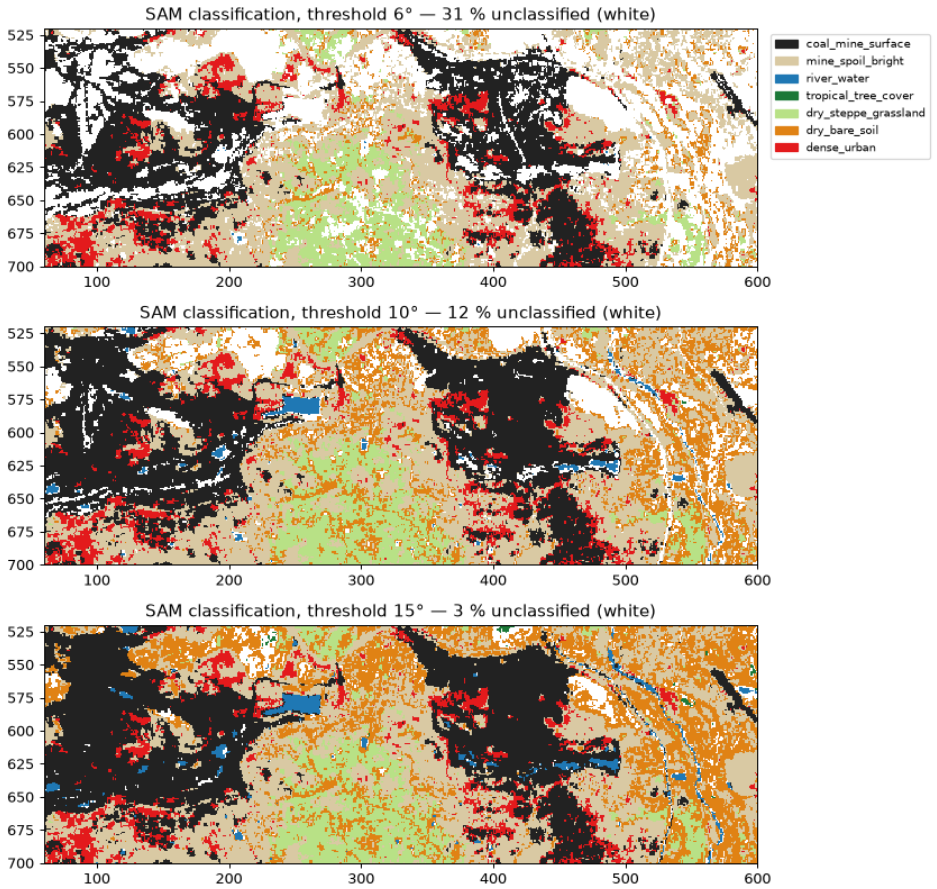
Pairwise SAM between the 19 course library spectra

- Small angles (dark cells) mean confusion is inevitable
- 2.83° dense urban ↔ bright industrial roof
- 2.91° sabkha ↔ desert sand
- 4.45° tropical tree cover ↔ sugarcane
- Fixes: texture, context, or a **time series** — not a better angle measure

SAM classification in three steps

- **1. Rule images** — one angle map per reference material (dark = similar)
- **2. Minimum-angle rule** — assign each pixel to its closest reference
- **3. Threshold** — if even the best angle is large, label the pixel *unclassified*
- Unclassified means: the material is not in the library, or the pixel is a mixture
- At 30 m, **most pixels are mixtures** — expect a large unclassified fraction

Minimum-angle classification of the mine



SAM classification, Kusmunda window (180 × 540 px)

- Works where references came from **this** scene: coal and spoil
- Farmland is forced into 'soil' or 'steppe' — imported references are weaker
- Set thresholds **per class** from rule-image histograms, not one global value
- Always **report the unclassified fraction** — it is a quality indicator
- SAM and SID agree on **83.9 %** of pixels; they differ where classes are close

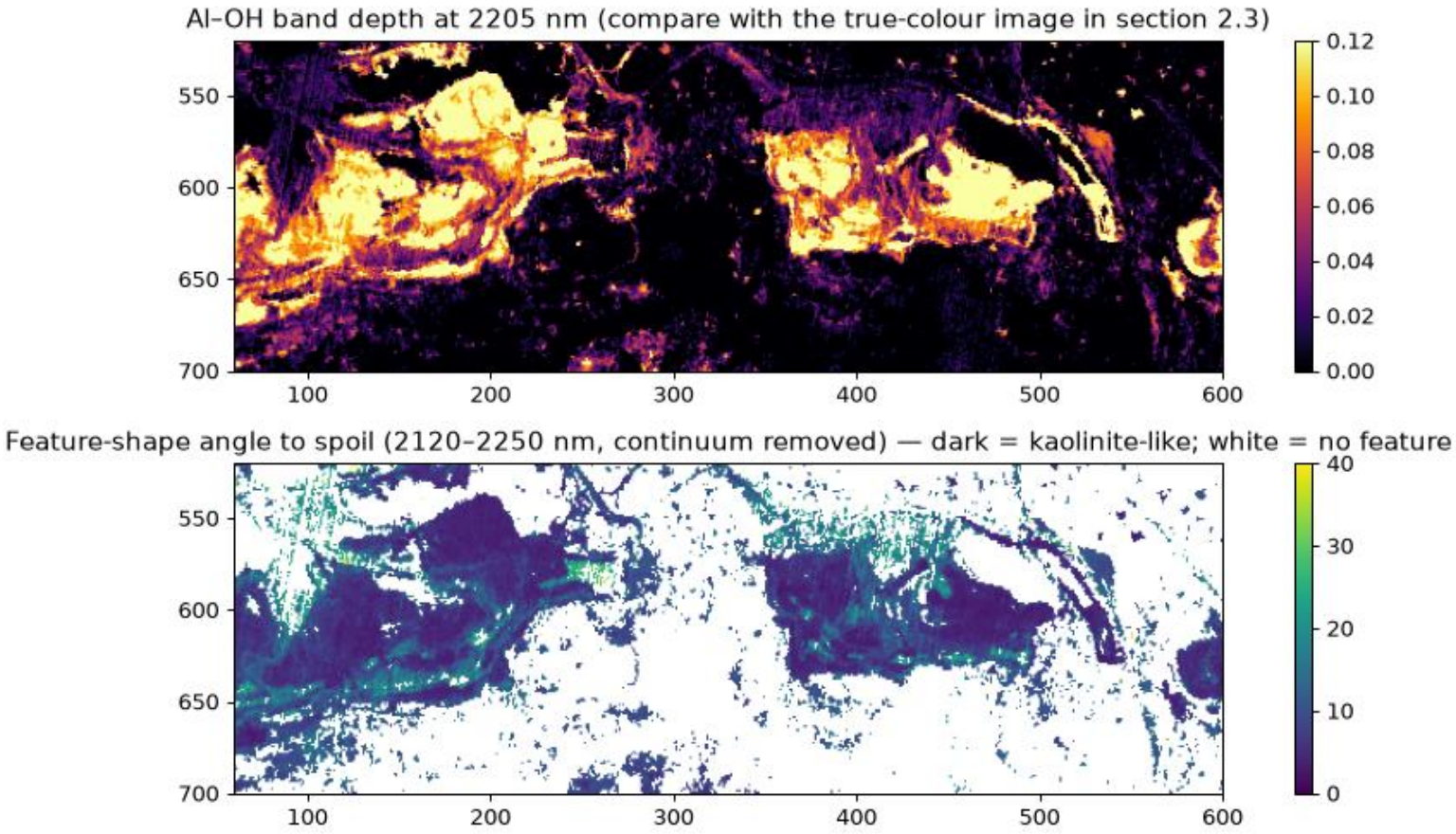
3. Matching features, not spectra

Continuum removal

Why geologists do not use whole-spectrum SAM

- Whole-spectrum SAM is dominated by broad shape: brightness slope, the vegetation red edge
- **Continuum removal** (Clark & Roush, 1984) divides the spectrum by its upper convex hull
- Absorption features become dips below 1.0, comparable between bright and dark materials
- Clays show **Al–OH near 2200 nm**; kaolinite has a doublet at ~2165 and ~2205 nm
- Match the *feature shape* with SAM restricted to 2120–2250 nm
  - Check against residual CO_2 (~2010, 2060 nm) and water vapour (~1900 nm) artefacts

Feature depth and feature shape at the mine



2205 nm band depth and continuum-removed shape matching, Kusmunda

4. Target detection

Finding what is rare and partly hidden

Why similarity measures fail for small targets

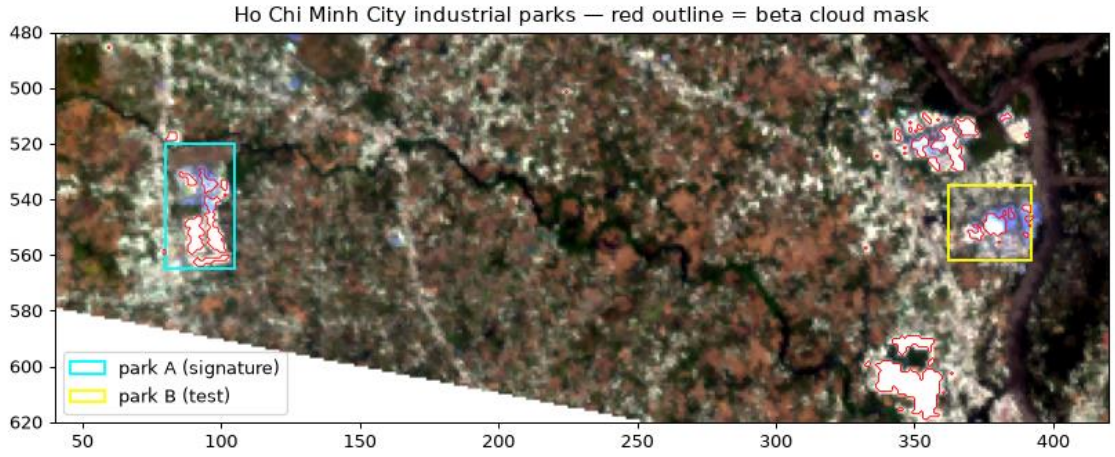
- SAM compares the **whole** pixel with the reference
- If a target fills 10 % of a 30 m pixel, 90 % of the signal is background
- The angle to the target stays large, so the pixel is never selected
- **Detectors** instead model the background statistically
- They look for the *direction* in which the target pushes a pixel away from the background
  - Assumption: targets are **rare**, so scene statistics ≈ background statistics

Three detectors

Detector	Score	Notes
Matched filter	$s^T\Sigma^{-1}(x - \mu) / s^T\Sigma^{-1}s$	≈ target fraction under linear mixing
ACE	$(s^T\Sigma^{-1}(x - \mu))^2 / (s^T\Sigma^{-1}s \cdot (x-\mu)^T\Sigma^{-1}(x-\mu))$	$\cos^2$ of the whitened angle; range 0–1
CEM	w^Tx, $w = R^{-1}t / (t^TR^{-1}t)$	Minimises output energy; uses correlation R

The target: blue metal roofs in Ho Chi Minh City

- Blue roofs reflect relatively more blue than red
- **113 target pixels** found in park A by blue/red ratio
- Split in half: **56** give the detector signature, **57** are kept for implanting
- Even the purest roof pixels are mixtures — the vegetation red edge is visible
- That is acceptable: the detector needs a *direction*, not a pure spectrum



Industrial park A, the signature source

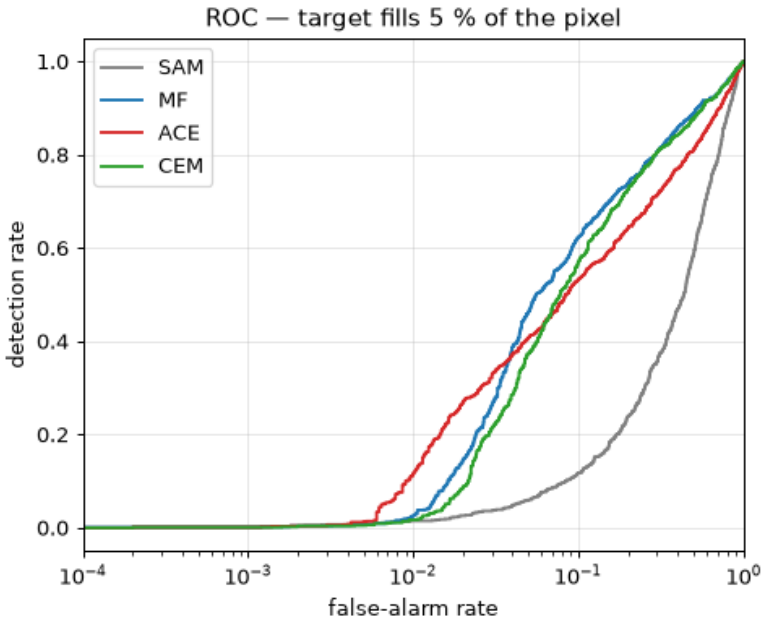
Background statistics and shrinkage

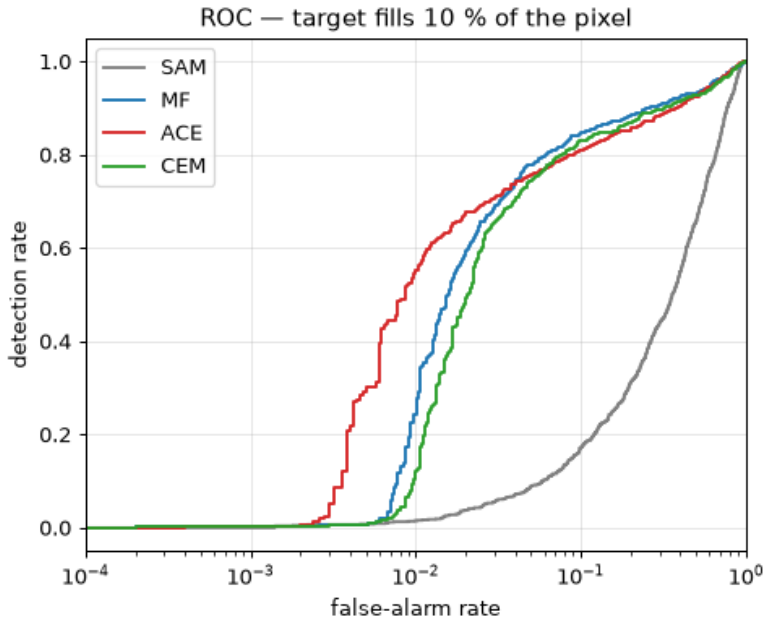
- μ and Σ come from all window pixels **outside** both parks
- With 350 bands, Σ is badly conditioned — condition number **1.6×10^{11}** without shrinkage
- Shrinkage λ on the diagonal: λ = 0.001 brings it to **2.8×10^{5}**
- Too little shrinkage over-fits noise; too much throws away the whitening that gives the power
- Tune λ on implanted targets — best AUC here was around λ = 0.001
  - Alternative: reduce dimensions with PCA or MNF first (notebook 13)

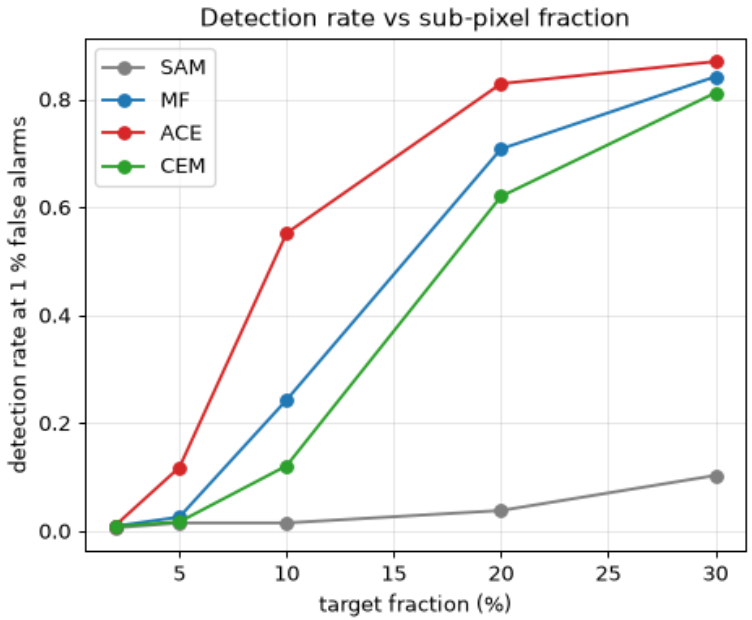
Benchmarking honestly: target implantation

- We rarely have ground truth for small targets, so we **make** some
- $x_{implant} = (1 - f) \cdot x_{background} + f \cdot t_{pixel}$, with sub-pixel fraction f
- t_{pixel} is a **random roof pixel from the held-out half** — real targets vary
- Score 5,000 clean background pixels and 1,000 implanted pixels with each detector
- Draw **ROC curves**: detection rate against false-alarm rate
  - Log-scale the false-alarm axis — operational use needs very low false-alarm rates

ROC curves by sub-pixel fill fraction







5,000 background vs 1,000 implanted pixels, Ho Chi Minh City

AUC, HIGHER IS BETTER

Area under the ROC curve by fill fraction

Fill	MF	CEM	ACE	SAM
2 %	0.676	0.669	0.581	0.537
5 %	0.828	0.815	0.772	0.562
10 %	0.913	0.902	0.906	0.612
20 %	0.937	0.929	0.937	0.722
30 %	0.938	0.930	0.944	0.790

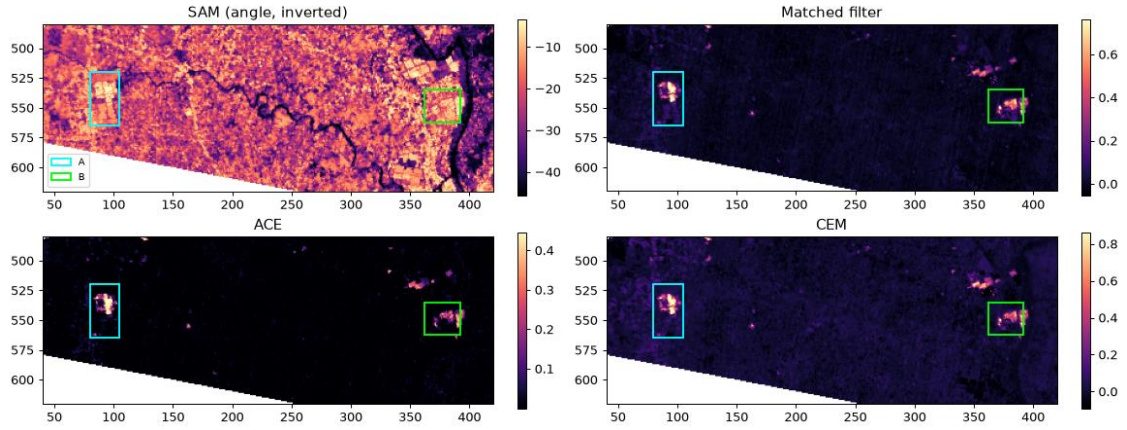
THE CAVEAT SLIDE

Read implantation results with care

- They are **optimistic**: linear mixing is assumed, with no misregistration or adjacency effects
- Implanted spectra come from the same park as the signature
- Treat AUC as an **upper bound** and as a way to *compare* detectors and settings
- The real value is relative: which detector, which λ, which band subset
- Then confirm on the real scene, and look at the map

On the real scene

- SAM lights up every bright, flat-spectrum surface in the city
- MF, ACE and CEM suppress the background and find **park B**
  - The signature came only from park A — this is genuine generalisation
- Faint vertical stripes in MF and CEM are **pushbroom detector striping**
  - Whitening amplifies tiny column-to-column calibration differences
- ACE above the 99.8th background percentile (0.223): **301 pixels**, 126 in A, 81 in B



SAM, MF, ACE and CEM over the same window

5. Spectral unmixing

From detection to fractions

The linear mixing model

- If materials sit side by side at sub-pixel scale, the spectrum is a weighted sum of **endmembers**
- $x = \Sigma\ a_i \cdot e_i + n$, with abundances a_i as area fractions
- **ANC**: $a_i \geq 0$ — no negative areas
- **ASC**: $\Sigma\ a_i = 1$ — the fractions cover the pixel
- Breaks down for **intimate mixtures** (mineral grains) and **canopy multiple scattering**
  - The residual map tells you where the model failed

Finding the endmembers

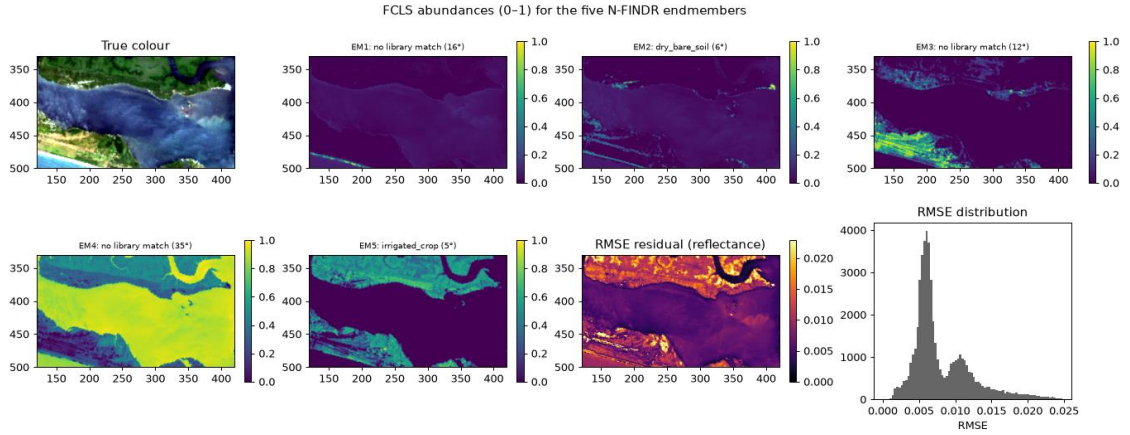
- With p endmembers the data occupy a **(p − 1)-dimensional simplex**
- Here the first 4 PCs hold **99.89 %** of the variance, so p = 5 is a reasonable choice
- **N-FINDR** (Winter, 1999): find the p pixels spanning the **largest simplex volume**
- Alternatives: **PPI** (Boardman, 1993) and **VCA** (Nascimento & Bioucas-Dias, 2005)
- Endmembers are the most **extreme** pixels, not necessarily the materials you care about
  - Breaking surf appeared as an endmember here — bright, flat, not in any library

Three ways to estimate abundances

Method	Negative fractions	Median sum	Median RMSE
Unconstrained LS	94.2 % of pixels	0.889	0.0031
NNLS ($a_i \geq 0$)	0 %	0.940	0.0065
FCLS ($a_i \geq 0$, $\Sigma = 1$)	0 %	1.000	0.0065

Abundance maps and the residual

- FCLS fractions can be read as **area percentages**
- The **residual map** is the diagnostic layer
- High residual = a material no endmember represents, or non-linear mixing
- Highest here on the **barrier island** (sand, palms, grass, fields in one pixel)
- And in **mangroves**, where a green-crop endmember fits the canopy shape badly
  - Remedies: more endmembers, MESMA (Roberts et al., 1998), non-linear models



Abundances and RMSE residual, Jiquilisco Bay

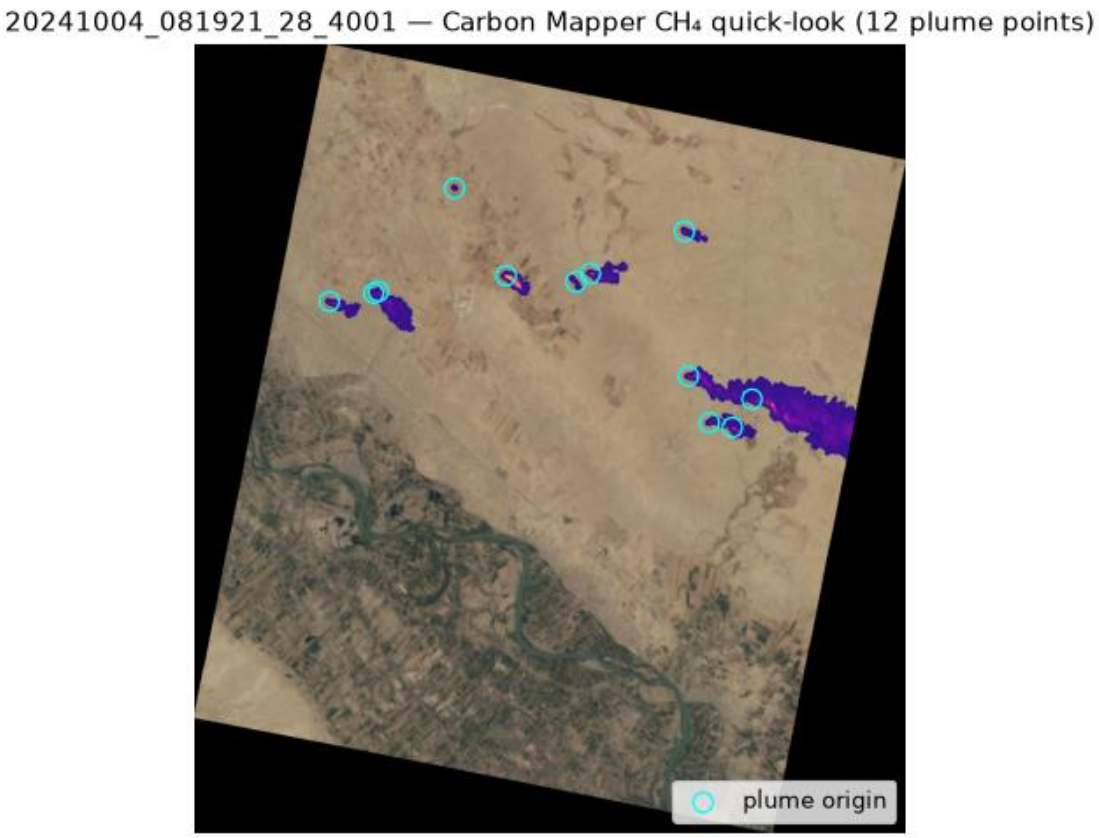
6. The methane case

Where the matched filter earns its keep

Methane retrieval is a matched filter on radiance

- CH_4 absorbs in narrow lines between **2200 and 2400 nm** — Tanager was designed for this
- Standard retrieval: Thompson et al. (2015); **MAG1C**, Foote et al. (2020); Carbon Mapper operationally
- Target is a **unit absorption spectrum**: $t = \mu \odot k$, with k from a radiative transfer model
- The output $\hat{\alpha}$ is the **methane enhancement** per pixel, in ppm·m
- **Why radiance?** Atmospheric correction treats gas absorption as atmosphere and partly removes the plume
  - So a reflectance-based band ratio is working on data the plume has been taken out of

Why notebook 03's MBSI is only a proxy



MBSI on methane-free reference spectra

- MBSI computed on **library spectra that contain no methane**
- Vegetation scores positive — leaf water, cellulose and lignin absorb near 2300 nm
- Carbonate- and sulphate-rich **sabkha** scores positive too
- A real plume adds only a few percent absorption — surface variability swamps it
- The matched filter whitens that variability away

Planet ships the operational product

- `ortho_radiance_hdf5` — the radiance you need for your own matched filter
- `ortho_ql_ch4` — a Carbon Mapper methane quick-look COG (qualitative concentration index)
- `ql_ch4_json` — plume origins as GeoJSON with emission rate, uncertainty and wind
- Example: Deir ez-Zor, Syria — **12 plumes**, roughly 170 to 3,200 kg/h, each with a quality flag
- Our Korba scene lists plume IDs, but its quick-look and plume file are empty
  - Check the quality flag: several plumes in that scene are marked *questionable*

PRACTICAL

Over to you

▶ `14_material_discrimination_target_detection.ipynb` · 150 min

- Build SAM rule images for the Kusmunda mine and set per-class thresholds
- Map the 2205 nm clay feature with continuum removal
- Run MF, ACE and CEM on the blue-roof target, and tune the shrinkage λ
- Reproduce the implantation experiment and the ROC curves
- Unmix the El Salvador estuary with N-FINDR and FCLS, then inspect the residual map

KEY TAKEAWAYS

Recap

- **SAM and SID** compare shape, not brightness — check class separability before promising a map
- Report the **unclassified fraction**; at 30 m most pixels are mixtures
- **Continuum removal** isolates diagnostic features and beats whole-spectrum matching for minerals
- **Detectors** (MF, ACE, CEM) model the background and find **sub-pixel** targets that SAM misses
- **Shrinkage** of the covariance is not optional at 350 bands
- **Implantation** gives you an honest, if optimistic, way to compare detectors
- **FCLS** abundances are area fractions; the **residual map** shows where the model fails
- Methane needs a **matched filter on radiance** — band ratios respond to the surface

Next: Session 4 — Classification approaches for hyperspectral data

References and further reading

Kruse, F.A. et al. (1993). The Spectral Image Processing System (SIPS). RSE 44, 145–163. (SAM)

Chang, C.-I. (2000). An information-theoretic approach to spectral variability, similarity and discrimination. IEEE Trans. Inf. Theory 46, 1927–1932. (SID)

Clark, R.N. & Roush, T.L. (1984). Reflectance spectroscopy: quantitative analysis techniques. JGR 89, 6329–6340. (Continuum removal)

Harsanyi, J.C. (1993). Detection and classification of subpixel spectral signatures in hyperspectral image sequences. PhD thesis, UMBC. (CEM)

Kraut, S. & Scharf, L.L. (1999). The CFAR adaptive subspace detector is a scale-invariant GLRT. IEEE Trans. Signal Processing 47, 2538–2541. (ACE)

Winter, M.E. (1999). N-FINDR: an algorithm for fast autonomous spectral endmember determination. Proc. SPIE 3753, 266–275.

Nascimento, J.M.P. & Bioucas-Dias, J.M. (2005). Vertex component analysis. IEEE TGRS 43, 898–910.

Heinz, D.C. & Chang, C.-I. (2001). Fully constrained least squares linear spectral mixture analysis. IEEE TGRS 39, 529–545.

Roberts, D.A. et al. (1998). Mapping chaparral in the Santa Monica Mountains using multiple endmember spectral mixture models. RSE 65, 267–279. (MESMA)

Thompson, D.R. et al. (2015). Real-time remote detection and measurement for airborne imaging spectroscopy. AMT 8, 4383–4397.

Foote, M.D. et al. (2020). Fast and accurate retrieval of methane concentration from imaging spectrometer data using sparsity prior. IEEE TGRS 58, 6480–6492. (MAG1C)